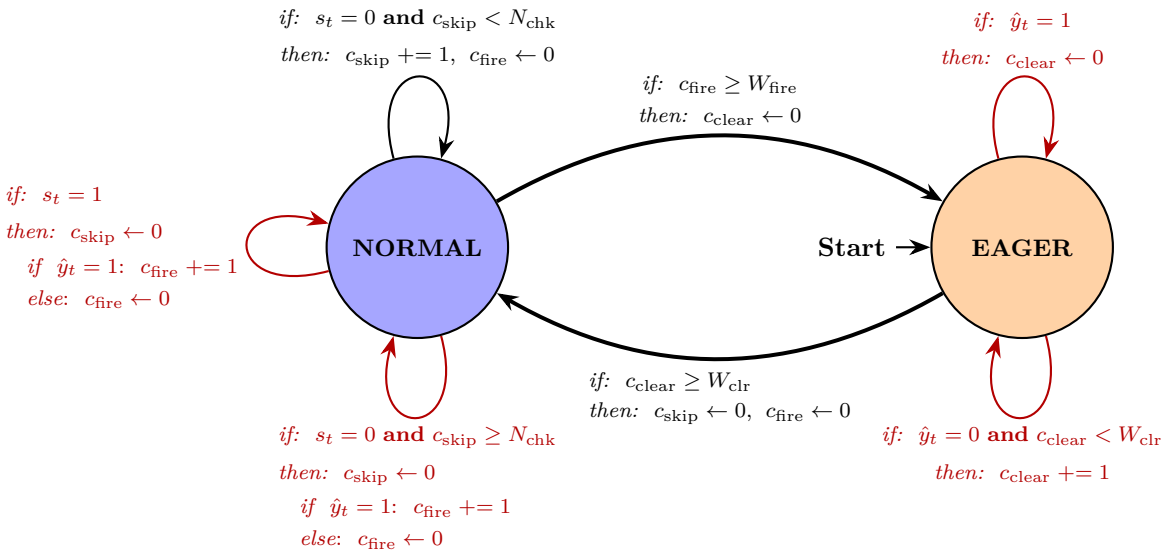




Symbol / Mode	Description
● NORMAL	Skip module $\mathcal{S}$ is active : static frames are skipped; $\mathcal{M}$ is invoked only when motion is detected ($s_t=1$) or $c_{\text{skip}} \geq N_{\text{chk}}$
● EAGER	Skip module $\mathcal{S}$ is bypassed : $\mathcal{M}$ runs on <i>every</i> frame to prevent missed detections. <i>Each video initializes in this mode.</i>
→	$\mathcal{M}(f_t)$ is invoked on this transition (either motion detected or forced call)
$\rightarrow$	$\mathcal{M}(f_t)$ is not invoked (frame skipped)
$s_t \in \{0, 1\}$	Output of $\mathcal{S}$: $s_t=0$ no motion (skip); $s_t=1$ motion detected (run $\mathcal{M}$)
$\hat{y}_t \in \{0, 1\}$	Output of $\mathcal{M}$: $\hat{y}_t=1$ fire/smoke; $\hat{y}_t=0$ negative
c_{skip}	Consecutive skipped frames; reset to 0 on any $\mathcal{M}$ call
c_{fire}	Consecutive fire/smoke predictions in NORMAL; reset to 0 on any negative or on exiting EAGER mode
c_{clear}	Consecutive negative predictions in EAGER; reset to 0 on any positive prediction or on entering EAGER mode
W_{fire}	Streak threshold to enter EAGER
W_{clr}	Streak threshold to exit EAGER
N_{chk}	Max consecutive skips before forced $\mathcal{M}$ call